Problem 1

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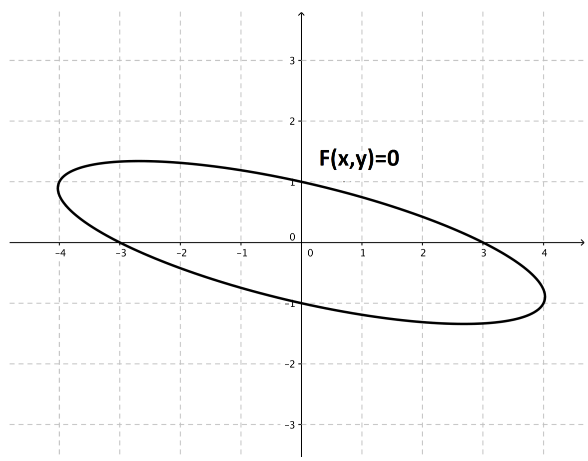
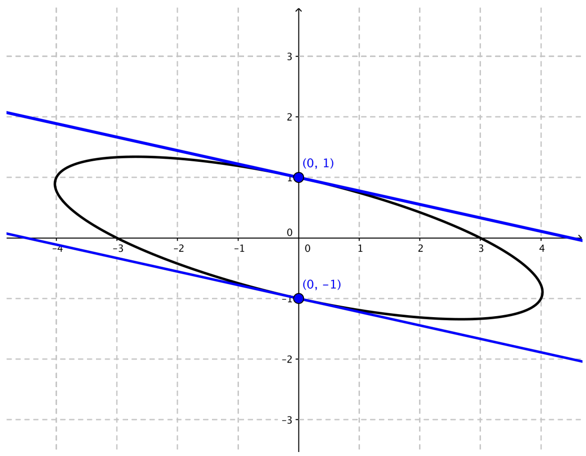
Problem 1 *On the graph below, sketch the tangent lines at $x = 0$. Then, explain why both the x-coordinate and the y-coordinate are generally needed to find the slope of the tangent line at a point on the graph of an equation of the form $F(x, y) = 0$*



Explanation. Given the equation $F(x, y) = 0$, its graph may not pass the "vertical line test". As illustrated in the figure below, fixing a value for x (in this case $x = 0$) will not typically specify a unique point on the graph because there may be more than one corresponding y-value. That is why you need to specify both the x and the y coordinate, to make sure that you are giving the coordinates for a unique point on the graph. In this case, there are two tangent lines to the curve at $x = 0$. One passes through the point $(0, 1)$ and the other through $(0, -1)$.


(0, 1)
(0, −1)
0
−4
−3
−2
−1
0
1
2
3
4
3
2
1
−1
−2
−3